Jawad Mohammed

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PROFESSIONAL SUMMARY

Software engineer with **6+ years** building fault-tolerant, high-availability distributed systems in **Java, Spring Boot, Kafka and Kubernetes**. At Infosys (client: Verizon IoT), migrated COBOL workloads to Java microservices on Kafka. At Thomson Reuters, moved 10+ Jenkins CI pipelines to AWS and kept production at **zero critical incidents for 15 months**. Now focused on platform engineering and reliability: GitOps with Terraform and Argo CD, chaos testing and observability.

EDUCATION

MS, Computer Science and Engineering | University of North Texas, Denton, TX | Expected Dec 2026

M.Tech, Computer Science and Engineering | Vasavi College of Engineering, India | Aug 2019

B.E., Information Technology | Deccan College of Engineering and Technology, India | Jul 2017

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, Bash

Platform and DevOps: Linux, Kubernetes, Docker, Terraform, Argo CD, GitOps, Gateway API, Jenkins, GitHub Actions, CI/CD, Maven, Gradle

Cloud: AWS (S3, EC2, DynamoDB, Lambda, Step Functions); EKS and IRSA defined in Terraform

Reliability: Prometheus, Grafana, load and chaos testing, autoscaling, root cause analysis

Backend and data: Spring Boot, Spring Batch, Spring Cloud, JPA/Hibernate, Kafka, Kafka Streams, PostgreSQL, Redis, MongoDB

Architecture: Distributed systems, microservices, event-driven architecture, REST APIs, system design, data structures and algorithms

AI-assisted development: GitHub Copilot for coding, testing, debugging and documentation, with tests and code review as the check

PROFESSIONAL EXPERIENCE

Senior Associate Consultant | Infosys (Client: Verizon IoT), India | Apr 2023 - Nov 2024

- Helped design and roll out **4 Spring Boot microservices** for IoT event ingestion during a migration off legacy COBOL, raising throughput **20%** at about 10K events per cycle.
- Owned the **Kafka Streams** integration, sustaining **10K+ events per minute** at peak device load with reliable, low-latency delivery.
- Refactored COBOL business logic into object-oriented Java service layers, removing procedural bottlenecks.
- Built fault-tolerant **Spring Batch** pipelines that kept **100% data integrity**, so a single bad record never stalled a job.

Tech: Java, Spring Boot, Kafka, Kafka Streams, PostgreSQL, Redis, Kubernetes, Spring Batch, Docker

Tech Intern, Associate Developer QA, Developer QA | Thomson Reuters, India | Sep 2018 - Mar 2023

- Served on the **production on-call rotation** and kept production at **zero critical incidents for 15 consecutive months** during a staffing shortage.
- Supported **Linux-based QA, UAT, SAT and production environments** through releases, deployments and incident triage.
- Drove root cause analysis and preventive monitoring on production incidents, reducing repeat outages **15%**.
- Led the migration of **10+ Jenkins CI pipelines to AWS** with a partner team (Cognizant), keeping releases on schedule and cutting release cycle time **30%**.
- Built an API validation framework enforced in CI/CD during the Spring Boot migration, cutting defect rates **30%**.
- Benchmarked AWS S3 and DynamoDB against PostgreSQL at 10K+ transactions, validating **25% higher throughput**.
- Automated **40%** of manual regression testing with JUnit, Mockito and Cucumber, keeping **100%** statutory compliance accuracy.
- Owned features end to end, from product-owner intake and functional specs to engineering handoff and client demos.
- Promoted from intern to Associate Developer in **10 months** on the Financial Reporting platform.

Tech: Java, Spring Boot, PostgreSQL, Redis, JPA/Hibernate, REST APIs, AWS, Jenkins, Docker, JUnit, Mockito, Cucumber

Graduate Teaching Assistant | University of North Texas, Denton, TX | Aug 2025 - Present

- Support a 100-student Foundations of Computing course through weekly sessions, office hours, grading and proctoring.

PROJECTS

EKS GitOps Platform | *Terraform, Kubernetes, Argo CD, Gateway API, autoscaling*

- Defined VPC, EKS and IRSA in **Terraform** (12 offline tests); **Argo CD** deploys every change from Git.
- On a local cluster: **14.6 s** from git push to all pods updated, **1.4 s** rollback with git revert, and **99.997%** request success during releases.
- Scaled **2 to 6 pods in 69 s** under a 3x spike and cut fast 503 errors from **21% to 2.7%**.

IoT Telemetry Pipeline | *Kafka, stream processing, PostgreSQL, Redis*

- Event pipeline with real-time device status and offline detection from missing heartbeats.
- Sustained **500 events/sec** at about 50 ms p95, and **2,000 events/sec** with no loss.
- Lost **zero acknowledged events across 2.5 million** while killing a broker, consumer, processor and ingest node.

PulseLink Real-Time Alerting | *Java, WebSockets, Prometheus, Grafana, Docker*

- Streaming patient alerts with escalation, an audit trail and authenticated WebSocket dashboards (47 tests).
- Load testing exposed a latency bug; the fix cut p99 from **1.5 s to 0.13 s**.
- Raised throughput **24%**, cut backlog alert p95 from **8.2 s to 0.5 s**, and held **154 ms** p95 at 3x traffic with no alerts lost.

PUBLICATIONS

Mohammed, J., and Hossain, G. "Detecting Race Conditions in FHIR-Based Healthcare Interoperability: A Formal Graph-Theoretic Approach." Accepted for publication, IIETA, 2026. Preprint: arXiv:2604.03043.

- Graph-based detector (FRAG) for three FHIR race-condition classes with formal proofs: **90.0% F1 vs 25.5%** for a baseline on 1,500 synthetic logs, scaling near-linearly to **250,000 events**.